

2023 AMC 10A Problem 18 - Solution

Review the full statement and step-by-step solution for 2023 AMC 10A Problem 18. Great practice for AMC 10, AMC 12, AIME, and other math contests

2023 AMC 10A Problem 18

Problem:

A rhombic dodecahedron is a convex polyhedron where each of the 12 faces is a rhombus, and all of the faces are congruent to each other. The number of edges that meet at a vertex is either 3 or 4, depending on the vertex. What is the number of vertices at which exactly 3 edges meet?

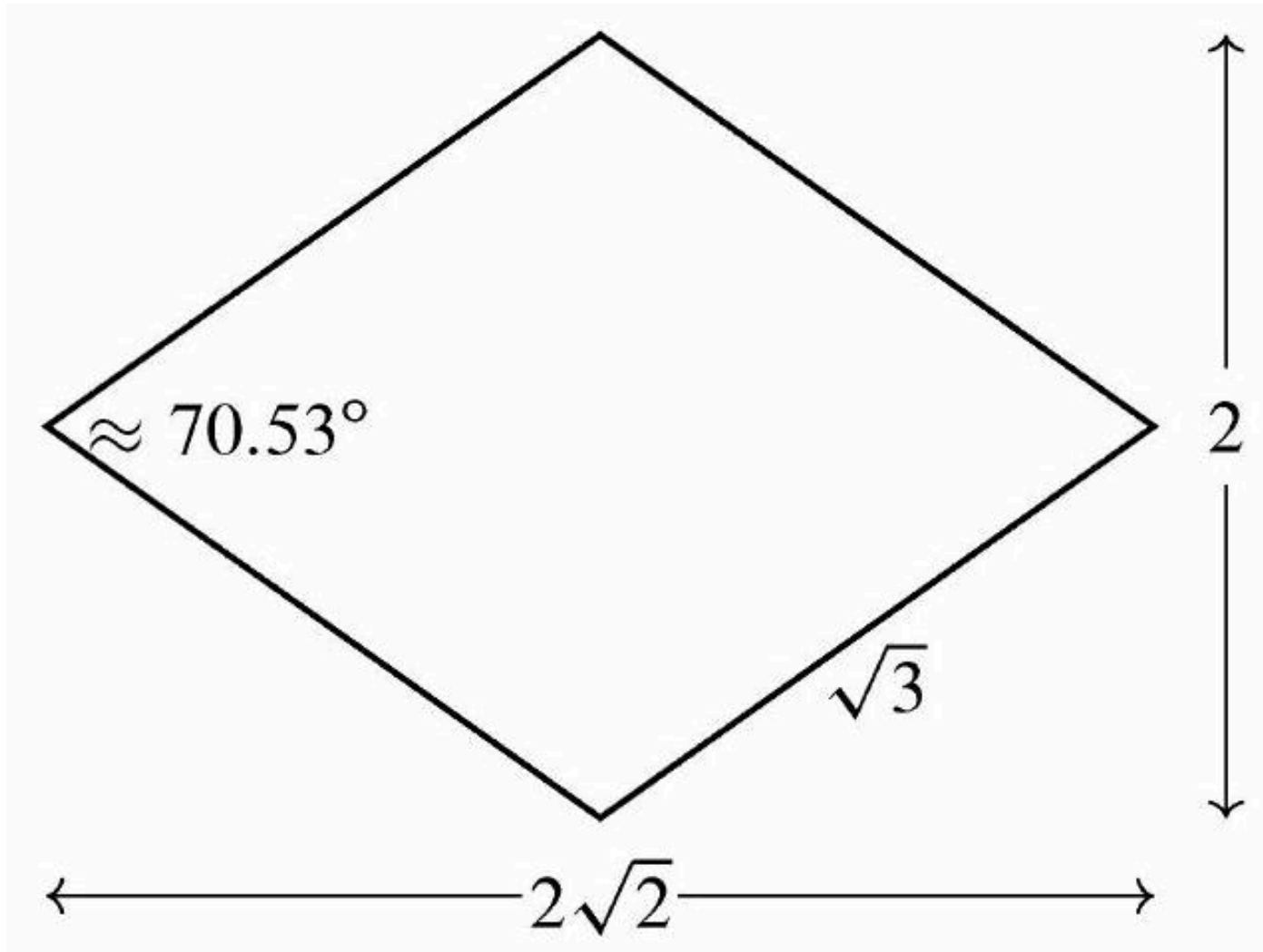
Answer Choices:

- A. 5
- B. 6
- C. 7
- D. 8
- E. 9

Solution:

Let V , E , and F denote the number of vertices, edges, and faces of the rhombic dodecahedron, respectively. Thus $F = 12$. Each face has 4 edges, and each edge belongs to exactly 2 faces, so $E = \frac{4 \cdot 12}{2} = 24$. Euler's Formula, $V - E + F = 2$, gives $V = 14$. Let V_3 denote the number of vertices at which 3 edges meet, and let V_4 the number of vertices at which 4 edges meet. Thus $V_3 + V_4 = 14$. The quantity $3V_3 + 4V_4$ counts each edge twice, once for each of its endpoints, so $3V_3 + 4V_4 = 48$. Solving this system of equations yields $V_3 = \text{(D)} \boxed{8}$ and $V_4 = 6$.

Note: In order to form a rhombic dodecahedron, it is necessary that each rhombus has the dimensions shown below.



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